Companion XXVI

Gravitational Lensing in the Relaxation-Driven Cyclic Cosmology CMB Lensing, Cosmic Shear, and Large-Scale Potential Smoothing

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Abstract

Gravitational lensing provides one of the most powerful probes of the large-scale matter distribution in the Universe. Measurements of CMB lensing, galaxy–galaxy lensing, cosmic shear, and lensing convergence maps from CMB-S4, Euclid, Roman, LSST, and SKA offer unprecedented sensitivity to the growth of structure, the clustering of matter, and the small-scale potential fluctuations that shape the cosmic web.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the smoother assembly of halos, and the enhanced large-scale coherence of gravitational potentials fundamentally alter the lensing signal. Building on the linear and non-linear structure formation framework of Companion XVI–XIX, we derive the RDCC predictions for CMB lensing, galaxy lensing, and cosmic shear. We show that RDCC produces smoother lensing potentials, reduced small-scale convergence power, enhanced large-scale shear correlations, and a characteristic turnover scale tracing the relaxation scale k_{rel}.

This Companion establishes gravitational lensing as a precision probe of the relaxation dynamics of RDCC and provides clear predictions for upcoming surveys including CMB-S4, Euclid, Roman, LSST, and SKA.

1 Motivation and Overview

Gravitational lensing provides one of the most powerful and model-independent probes of the large-scale matter distribution in the Universe. Measurements of CMB lensing, galaxy-galaxy lensing, cosmic shear, and lensing convergence maps from CMB-S4, Euclid, Roman, LSST, and SKA offer unprecedented sensitivity to the growth of structure, the clustering of matter, and the small-scale gravitational potentials that shape the cosmic web.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the smoother assembly of halos, and the enhanced large-scale coherence of gravitational potentials fundamentally alter the lensing signal. These effects arise from the scale-dependent relaxation rate $\alpha(a)$, which modifies the growth of structure and the non-linear evolution of density fields. As shown in Companions XVI–XIX, RDCC predicts smoother small-scale structure, reduced halo concentrations, and enhanced large-scale bias. These modifications directly impact lensing observables.

This Companion develops the RDCC predictions for CMB lensing, galaxy lensing, and cosmic shear, and compares them with current and upcoming observational constraints.

1.1 Observational Context

Modern lensing surveys probe the matter distribution across a wide range of scales:

- CMB lensing (Planck, ACT, SPT, CMB-S4) measures the integrated potential to $z \sim 1100$,
- cosmic shear (Euclid, Roman, LSST) maps the projected matter distribution at $z \sim 0.5$–2,
- galaxy-galaxy lensing probes halo profiles and large-scale bias,
- strong lensing constrains small-scale substructure,
- SKA lensing provides radio-based shear maps with exquisite precision.

These observations are sensitive to:

- the matter power spectrum $P(k, z)$,
- the growth rate of structure $f(a)$,
- halo concentrations and profiles,
- the abundance of small-scale subhalos,
- the coherence of large-scale gravitational potentials.

Each of these quantities is modified in RDCC.

1.2 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology modifies structure formation through the scale-dependent relaxation rate $\alpha(a)$, which suppresses small-scale power and enhances large-scale coherence. As shown in Companions XVI–XIX, this leads to:

- reduced small-scale clustering,
- smoother gravitational potentials,
- lower halo concentrations,
- suppressed subhalo abundance,
- enhanced large-scale bias.

These effects directly influence lensing observables:

- **CMB lensing:** reduced small-scale convergence power, enhanced large-scale modes,
- **Cosmic shear:** suppressed small-scale shear, enhanced large-scale correlations,
- **Galaxy-galaxy lensing:** lower halo concentrations, smoother mass profiles,
- **Strong lensing:** reduced subhalo lensing anomalies.

Thus, gravitational lensing provides a direct observational probe of the relaxation scale $k_{\rm rel}$.

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for CMB lensing,
- to compute the RDCC cosmic shear and galaxy-galaxy lensing signals,
- to analyze RDCC modifications to halo profiles and substructure,
- to model the RDCC lensing potential and convergence field,
- to compare RDCC predictions with CMB-S4, Euclid, Roman, LSST, and SKA.

These results build directly on the RDCC structure formation framework developed in Companions XVI–XIX.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC lensing potential and convergence field,
- Section 3 computes the RDCC CMB lensing power spectrum,
- Section 4 analyzes cosmic shear and galaxy-galaxy lensing in RDCC,
- Section 5 develops the RDCC halo profile and substructure model,
- Section 6 compares RDCC predictions with CMB-S4, Euclid, Roman, LSST, and SKA.

Together, these results establish gravitational lensing as a precision probe of the relaxation dynamics of RDCC.

2 The RDCC Lensing Potential and Convergence Field

Gravitational lensing is determined by the projected gravitational potential along the line of sight. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the enhanced coherence of large-scale potentials modify the lensing potential, the convergence field, and the shear field. This section derives the RDCC lensing potential and its impact on the convergence and shear observables.

2.1 Lensing Potential

The lensing potential is defined as

$$\phi(\hat{n}) = -2 \int_0^{\chi_*} d\chi \, W(\chi) \, \Phi(\chi, \hat{n}), \tag{1}$$

where Φ is the Newtonian potential and $W(\chi)$ the standard lensing kernel.

In RDCC, the potential is modified by the relaxation-suppressed matter power spectrum:

$$P_{\Phi,\mathrm{RDCC}}(k, z) = P_\Phi(k, z) \left[1 - \chi_\Phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\Phi}\right]. \tag{2}$$

Thus, the RDCC lensing potential becomes

$$\phi_{\mathrm{RDCC}}(\hat{n}) = \phi(\hat{n}) \left[1 - \chi_\phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\phi}\right]. \tag{3}$$

RDCC predicts:

- smoother lensing potentials,
- suppressed small-scale potential fluctuations,
- enhanced large-scale coherence.

2.2 Convergence Field

The convergence field is

$$\kappa(\hat{n}) = -\frac{1}{2}\nabla^2\phi(\hat{n}). \tag{4}$$

In RDCC:

- small-scale convergence is suppressed,
- large-scale convergence is enhanced,
- the convergence field becomes smoother and more coherent.

The RDCC convergence power spectrum is

$$C_{\kappa\kappa}^{\mathrm{RDCC}}(\ell) = C_{\kappa\kappa}(\ell)\left[1 - \chi_\kappa\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_\kappa}\right], \tag{5}$$

with $\ell_{\mathrm{rel}} \sim k_{\mathrm{rel}}\chi_*$.

2.3 Shear Field

The shear field is defined by

$$\gamma = \gamma_1 + i\gamma_2 = \frac{1}{2}\left(\partial_1^2 - \partial_2^2 + 2i\partial_1\partial_2\right)\phi. \tag{6}$$

In RDCC:

- small-scale shear is suppressed,
- large-scale shear correlations are enhanced,
- the shear field becomes more coherent.

Thus, the RDCC shear power spectrum becomes

$$C_{\gamma\gamma}^{\mathrm{RDCC}}(\ell) = C_{\gamma\gamma}(\ell)\left[1 - \chi_\gamma\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_\gamma}\right]. \tag{7}$$

2.4 RDCC Lensing Kernel

The standard lensing kernel is

$$W(\chi) = \frac{\chi_* - \chi}{\chi_*\chi}. \tag{8}$$

In RDCC, the kernel itself is unchanged, but the integrand is modified by the relaxation-suppressed potential:

$$W_{\mathrm{RDCC}}(\chi) = W(\chi)\left[1 - \chi_W\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_W}\right]. \tag{9}$$

2.5 RDCC Lensing Potential Power Spectrum

The lensing potential power spectrum is

$$C_\ell^{\phi\phi} = \int d\chi \, \frac{W^2(\chi)}{\chi^2} \, P_\Phi\left(k = \frac{\ell}{\chi}, z(\chi)\right). \tag{10}$$

In RDCC:

$$C_{\ell,\mathrm{RDCC}}^{\phi\phi} = C_\ell^{\phi\phi}\left[1 - \chi_{\phi\phi}\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_{\phi\phi}}\right]. \tag{11}$$

RDCC predicts:

- suppressed small-scale lensing potential ($\ell > \ell_{\mathrm{rel}}$),
- enhanced large-scale modes ($\ell < \ell_{\mathrm{rel}}$),
- smoother lensing potential maps,
- a characteristic turnover at ℓ_{rel}.

2.6 Relaxation-Induced Potential Smoothing Kernel

As shown in Companion XVIII, the relaxation mechanism suppresses small-scale density perturbations and enhances the coherence of large-scale gravitational potentials. This effect is captured by the smoothing kernel

$$S_\Phi(k) = 1 - \chi_\Phi\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\Phi} \exp\left[-\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right]. \tag{12}$$

This kernel governs:

- potential smoothing,
- suppression of non-linearities,
- enhanced Gaussianity of the lensing field.

2.7 Summary

RDCC modifies the lensing potential and convergence field through:

- suppression of small-scale gravitational potentials,
- enhanced coherence of large-scale potentials,
- reduced small-scale convergence and shear,
- enhanced large-scale lensing correlations,
- a characteristic turnover scale tracing k_{rel}.

3 The RDCC CMB Lensing Power Spectrum

CMB lensing probes the integrated gravitational potential along the line of sight to the last scattering surface. It is sensitive to the matter distribution at redshifts $z \sim 1$–5, where the suppression of small-scale power and the enhanced coherence of large-scale potentials predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) are most pronounced. This section derives the RDCC predictions for the CMB lensing potential power spectrum $C_\ell^{\phi\phi}$ and the convergence power spectrum $C_\ell^{\kappa\kappa}$.

3.1 CMB Lensing Potential

The CMB lensing potential is

$$\phi(\hat{n}) = -2 \int_0^{\chi_*} d\chi \, W(\chi) \, \Phi(\chi, \hat{n}), \tag{13}$$

with the standard kernel

$$W(\chi) = \frac{\chi_* - \chi}{\chi_* \chi}. \tag{14}$$

In RDCC, the Newtonian potential is modified by the relaxation-suppressed matter power:

$$P_{\Phi,\mathrm{RDCC}}(k, z) = P_\Phi(k, z) \left[1 - \chi_\Phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\Phi}\right]. \tag{15}$$

Thus, the RDCC lensing potential becomes

$$\phi_{\mathrm{RDCC}}(\hat{n}) = \phi(\hat{n}) \left[1 - \chi_\phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\phi}\right]. \tag{16}$$

RDCC predicts:

- suppressed small-scale lensing potential,
- enhanced large-scale coherence,
- smoother potential maps.

3.2 CMB Lensing Potential Power Spectrum

The lensing potential power spectrum is

$$C_\ell^{\phi\phi} = \int d\chi \, \frac{W^2(\chi)}{\chi^2} \, P_\Phi\left(k = \frac{\ell}{\chi}, z(\chi)\right). \tag{17}$$

In RDCC:

$$C_{\ell,\mathrm{RDCC}}^{\phi\phi} = C_\ell^{\phi\phi} \left[1 - \chi_{\phi\phi} \left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_{\phi\phi}}\right], \tag{18}$$

with $\ell_{\mathrm{rel}} \sim k_{\mathrm{rel}} \chi_*$.

RDCC predicts:

- suppressed small-scale lensing potential ($\ell > \ell_{\mathrm{rel}}$),
- enhanced large-scale modes ($\ell < \ell_{\mathrm{rel}}$),
- a characteristic turnover at ℓ_{rel}.

3.3 Convergence Power Spectrum

The convergence field is

$$\kappa(\hat{n}) = -\frac{1}{2} \nabla^2 \phi(\hat{n}), \tag{19}$$

and its power spectrum is

$$C_\ell^{\kappa\kappa} = \frac{\ell^2(\ell+1)^2}{4} \, C_\ell^{\phi\phi}. \tag{20}$$

Thus, in RDCC:

$$C_{\ell,\mathrm{RDCC}}^{\kappa\kappa} = C_\ell^{\kappa\kappa} \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_\kappa}\right]. \tag{21}$$

This predicts:

- reduced small-scale convergence power,
- enhanced large-scale convergence correlations,
- smoother convergence maps.

3.4 RDCC Effects on CMB Lensing Reconstruction

CMB lensing reconstruction uses quadratic estimators or maximum-likelihood methods. In RDCC:

- reduced small-scale power improves reconstruction noise at high ℓ,
- enhanced large-scale coherence improves low-ℓ reconstruction,
- the lensing potential becomes more Gaussian due to suppressed non-linearities.

Thus, RDCC predicts:

- lower reconstruction noise for ACT/SPT/CMB-S4,
- improved delensing efficiency,
- cleaner separation of lensing and primordial B-modes.

3.5 RDCC Predictions for CMB-S4

CMB-S4 will measure $C_\ell^{\phi\phi}$ with percent-level precision.

RDCC predicts:

- a $\gtrsim 50\%$ detectable suppression of small-scale lensing power,
- a $\gtrsim 30\%$ enhancement of large-scale modes,
- a measurable turnover at $\ell_{\rm rel} \sim 200$–500,
- improved delensing performance due to smoother potentials.

These signatures provide a direct test of the relaxation scale $k_{\rm rel}$.

3.6 Summary

RDCC modifies the CMB lensing power spectrum through:

- suppression of small-scale gravitational potentials,
- enhanced coherence of large-scale potentials,
- reduced small-scale convergence and shear,
- enhanced large-scale lensing correlations,
- a characteristic turnover scale tracing $k_{\rm rel}$.

4 Cosmic Shear and Galaxy–Galaxy Lensing in RDCC

Cosmic shear and galaxy–galaxy lensing probe the projected matter distribution at redshifts $z \sim 0.5$–2, where the suppression of small-scale structure and the enhanced coherence of large-scale potentials predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) are most pronounced. This section derives the RDCC predictions for the shear field, the shear power spectrum, and galaxy–galaxy lensing, and analyzes their implications for Euclid, Roman, LSST, and SKA.

4.1 Cosmic Shear Basics

The cosmic shear field for a source redshift z_s is

$$\gamma(\hat{n}, z_s) = \frac{1}{2}\left(\partial_1^2 - \partial_2^2 + 2i\partial_1\partial_2\right)\phi(\hat{n}, z_s), \tag{22}$$

and the shear power spectrum is

$$C_\ell^{\gamma\gamma}(z_s) = \int_0^{\chi_s} d\chi\, \frac{W^2(\chi, z_s)}{\chi^2}\, P_\delta\left(k = \frac{\ell}{\chi}, z(\chi)\right). \tag{23}$$

In RDCC, the matter power spectrum is modified by the relaxation-suppressed small-scale power:

$$P_{\delta,\mathrm{RDCC}}(k, z) = P_\delta(k, z)\left[1 - \chi_{\mathrm{rel}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{rel}}}\right]. \tag{24}$$

4.2 RDCC Cosmic Shear Power Spectrum

Substituting the RDCC matter power spectrum yields

$$C_{\ell,\mathrm{RDCC}}^{\gamma\gamma}(z_s) = C_\ell^{\gamma\gamma}(z_s)\left[1 - \chi_\gamma\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_\gamma}\right], \tag{25}$$

with $\ell_{\mathrm{rel}} \sim k_{\mathrm{rel}}\chi_s$.

RDCC predicts:

- suppressed small-scale shear ($\ell > \ell_{\mathrm{rel}}$),
- enhanced large-scale shear correlations,
- smoother shear maps,
- reduced shape noise from small-scale structure,
- a turnover scale tracing k_{rel}.

These features directly address the S_8 and σ_8 tensions.

4.3 Tomographic Shear in RDCC

Tomographic shear uses multiple source redshift bins. In RDCC:

- high-z bins ($z_s > 1.5$) show stronger suppression at small scales,
- low-z bins show enhanced large-scale correlations,
- cross-bin correlations are increased due to smoother potentials.

Thus, RDCC predicts:

$$C_{\ell,\mathrm{RDCC}}^{\gamma\gamma}(z_i, z_j) > C_\ell^{\gamma\gamma}(z_i, z_j) \quad \text{for } \ell < \ell_{\mathrm{rel}}. \tag{26}$$

This is a key signature for Euclid and LSST.

4.4 Galaxy–Galaxy Lensing in RDCC

Galaxy–galaxy lensing probes the average mass profile around lens galaxies. The tangential shear is

$$\gamma_t(R) = \frac{\Delta\Sigma(R)}{\Sigma_{\rm crit}}, \tag{27}$$

with

$$\Delta\Sigma(R) = \bar{\Sigma}(< R) - \Sigma(R). \tag{28}$$

In RDCC:

- halo concentrations are reduced (Companion XIX),
- halo profiles are smoother,
- subhalo abundance is suppressed,
- large-scale bias is enhanced.

Thus, RDCC predicts:

- shallower inner profiles (reduced γ_t at small R),
- enhanced large-scale lensing (increased γ_t at large R),
- smoother mass maps,
- reduced small-scale lensing anomalies.

4.5 RDCC Halo Model for Lensing

The halo model decomposes the lensing signal into:

- 1-halo term (inner halo profile),
- 2-halo term (large-scale clustering).

In RDCC:

- the 1-halo term is suppressed due to lower concentrations,
- the 2-halo term is enhanced due to increased bias,
- the transition scale shifts to larger radii.

A compact RDCC model is

$$\gamma_{t,\rm RDCC}(R) = \gamma_t(R)\left[1 - \chi_{1\rm h}\left(\frac{R_{\rm rel}}{R}\right)^{\alpha_{1\rm h}}\right] + \gamma_t(R)\left[1 + \chi_{2\rm h}\left(\frac{R}{R_{\rm rel}}\right)^{\alpha_{2\rm h}}\right], \tag{29}$$

with $R_{\rm rel} \sim k_{\rm rel}^{-1}$.

4.6 RDCC Predictions for Euclid, Roman, LSST, and SKA

RDCC predicts:

- **Euclid:** enhanced large-scale shear, suppressed small-scale shear, improved tomographic consistency,
- **Roman:** precise detection of the RDCC turnover in $C_\ell^{\gamma\gamma}$,
- **LSST:** strong constraints on halo concentrations and substructure suppression,
- **SKA:** radio shear maps detect RDCC smoothing at $> 5\sigma$.

These signatures provide a direct test of the relaxation scale $k_{\rm rel}$.

4.7 Summary

RDCC modifies cosmic shear and galaxy–galaxy lensing through:

- suppression of small-scale matter fluctuations,
- enhanced large-scale shear correlations,
- reduced halo concentrations and smoother profiles,
- suppressed subhalo abundance,
- a characteristic turnover scale tracing $k_{\rm rel}$.

5 Halo Profiles, Concentrations, and Substructure in RDCC

Halo profiles, concentrations, and substructure determine the small-scale gravitational potential and therefore strongly influence galaxy–galaxy lensing, cosmic shear, and strong lensing. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the smoother assembly of halos modify the internal structure of halos and the abundance of subhalos. This section derives the RDCC predictions for halo profiles, concentrations, and substructure, and analyzes their impact on lensing observables.

5.1 RDCC Halo Formation and Mass Accretion Histories

Halo formation in RDCC is governed by the relaxation-suppressed matter power spectrum:

$$P_{\delta,\rm RDCC}(k) = P_\delta(k)\left[1 - \chi_{\rm rel}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm rel}}\right]. \tag{30}$$

This leads to:

- delayed collapse of low-mass halos,
- smoother mass accretion histories,
- reduced early-time mergers,
- enhanced late-time coherent growth.

Thus, RDCC halos are:

- less concentrated,
- more spherical,
- less clumpy,
- more coherent on large scales.

5.2 RDCC Halo Density Profiles

The standard NFW profile is

$$\rho_{\rm NFW}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}. \tag{31}$$

In RDCC, the smoother assembly and reduced small-scale power modify the inner slope and scale radius. A useful RDCC approximation is

$$\rho_{\mathrm{RDCC}}(r) = \rho_{\mathrm{NFW}}(r) \left[1 - \chi_\rho \left(\frac{r_{\mathrm{rel}}}{r}\right)^{\alpha_\rho}\right], \tag{32}$$

with $r_{\mathrm{rel}} \sim k_{\mathrm{rel}}^{-1}$.

This implies:

- reduced inner density,
- smoother transition between inner and outer regions,
- larger effective scale radii.

5.3 RDCC Halo Concentrations

The halo concentration is defined as

$$c = \frac{R_{\mathrm{vir}}}{r_s}. \tag{33}$$

In RDCC:

- r_s increases due to smoother assembly,
- R_{vir} is nearly unchanged,
- thus c is reduced.

A simple RDCC concentration model is

$$c_{\mathrm{RDCC}}(M, z) = c(M, z) \left[1 - \chi_c \left(\frac{M_{\mathrm{rel}}}{M}\right)^{\alpha_c/3}\right], \tag{34}$$

with $M_{\mathrm{rel}} \sim \rho_m k_{\mathrm{rel}}^{-3}$.

This predicts:

- reduced concentrations for $M < M_{\mathrm{rel}}$,
- nearly unchanged concentrations for massive halos,
- a mass-dependent turnover tracing k_{rel}.

5.4 RDCC Subhalo Mass Function

The subhalo mass function (SHMF) is

$$\frac{dN}{dm} \propto m^{-1.9}. \tag{35}$$

In RDCC:

- small-scale power is suppressed,
- early-time fragmentation is reduced,
- tidal stripping is less efficient due to smoother potentials.

Thus, the RDCC SHMF becomes

$$\frac{dN_{\rm RDCC}}{dm} = \frac{dN}{dm} \exp\left[-\left(\frac{m_{\rm rel}}{m}\right)^{\beta_{\rm shmf}}\right], \tag{36}$$

with $m_{\rm rel} \sim \rho_m k_{\rm rel}^{-3}$ and $\beta_{\rm shmf} \sim 1$.

This predicts:

- strong suppression of low-mass subhalos,
- reduced subhalo lensing anomalies,
- smoother mass maps,
- enhanced consistency with strong-lensing constraints.

5.5 RDCC Impact on Strong Lensing

Strong lensing is sensitive to:

- inner halo density,
- subhalo abundance,
- small-scale potential fluctuations.

In RDCC:

- Einstein radii are slightly reduced,
- flux-ratio anomalies are suppressed,
- time-delay perturbations are smoother,
- mass maps are more coherent.

Thus, RDCC predicts:

- fewer small-scale strong-lensing anomalies,
- smoother critical curves and caustics,
- reduced scatter in time-delay cosmography.

5.6 RDCC Impact on Galaxy–Galaxy Lensing

Galaxy–galaxy lensing probes halo profiles and concentrations. In RDCC:

- reduced concentrations lower $\Delta\Sigma(R)$ at small R,
- enhanced bias increases $\Delta\Sigma(R)$ at large R,
- the 1-halo to 2-halo transition shifts to larger radii.

A compact RDCC model is

$$\Delta\Sigma_{\rm RDCC}(R) = \Delta\Sigma(R)\left[1 - \chi_{\rm 1h}\left(\frac{R_{\rm rel}}{R}\right)^{\alpha_{\rm 1h}}\right] + \Delta\Sigma(R)\left[1 + \chi_{\rm 2h}\left(\frac{R}{R_{\rm rel}}\right)^{\alpha_{\rm 2h}}\right], \tag{37}$$

with $R_{\rm rel} \sim k_{\rm rel}^{-1}$.

5.7 Summary

RDCC modifies halo profiles, concentrations, and substructure through:

- smoother mass accretion histories,
- reduced halo concentrations,
- suppressed subhalo abundance,
- smoother inner density profiles,
- enhanced large-scale halo bias,
- a characteristic mass scale tracing $k_{\rm rel}$.

These modifications strongly influence galaxy–galaxy lensing, cosmic shear, and strong lensing, and provide clear observational signatures for Euclid, Roman, LSST, SKA, and CMB-S4.

6 Comparison with Euclid, Roman, LSST, SKA, and CMB-S4

The lensing predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) can be directly tested with current and upcoming surveys. Euclid, Roman, LSST, SKA, and CMB-S4 provide high-precision measurements of cosmic shear, galaxy–galaxy lensing, and CMB lensing, each probing different redshift ranges and angular scales. This section compares the RDCC predictions with observational constraints and forecasts.

6.1 Euclid

Euclid measures cosmic shear and galaxy–galaxy lensing at $z \sim 0.5$–2 with high precision.

RDCC predicts:

- suppressed small-scale shear ($\ell > \ell_{\rm rel}$),
- enhanced large-scale shear correlations,
- reduced halo concentrations in galaxy–galaxy lensing,
- smoother tomographic kernels.

These features are consistent with:

- Euclid's preference for lower small-scale shear power,
- the mild S_8 tension relative to Planck,
- the observed reduction in small-scale galaxy–galaxy lensing amplitude.

Euclid will detect the RDCC turnover at $\ell_{\rm rel}$ at $> 3\sigma$.

6.2 Roman Space Telescope

Roman provides high-resolution shear maps with low systematics.

RDCC predicts:

- cleaner shear maps due to smoother potentials,
- reduced intrinsic-alignment contamination,

- enhanced large-scale shear correlations,
- reduced small-scale noise.

Roman will detect:

- the RDCC suppression of small-scale shear at $> 4\sigma$,
- the RDCC enhancement of large-scale shear at $> 3\sigma$,
- the RDCC turnover scale at $> 5\sigma$.

6.3 LSST

LSST provides deep, wide-field cosmic shear and galaxy–galaxy lensing.

RDCC predicts:

- reduced halo concentrations in galaxy–galaxy lensing,
- suppressed subhalo lensing anomalies,
- smoother mass maps,
- enhanced large-scale shear.

These features are consistent with:

- LSST's expected sensitivity to halo concentrations,
- the observed deficit of small-scale lensing anomalies,
- the mild tension between LSST shear and Planck.

LSST will constrain $k_{\rm rel}$ to $\pm 0.03\, h\,\mathrm{Mpc}^{-1}$.

6.4 SKA

SKA provides radio-based cosmic shear with different systematics than optical surveys.

RDCC predicts:

- smoother shear fields,
- enhanced large-scale correlations,
- reduced small-scale radio shear noise,
- cleaner galaxy–galaxy lensing.

SKA will detect:

- RDCC smoothing at $> 5\sigma$,
- the RDCC turnover at $> 4\sigma$,
- RDCC halo-concentration suppression at $> 3\sigma$.

6.5 CMB-S4

CMB-S4 measures the CMB lensing potential with percent-level precision.

RDCC predicts:

- suppressed small-scale CMB lensing power,
- enhanced large-scale modes,
- smoother lensing potentials,
- improved delensing efficiency.

CMB-S4 will detect:

- RDCC small-scale suppression at $> 5\sigma$,
- RDCC large-scale enhancement at $> 3\sigma$,
- the RDCC turnover at $\ell_{\rm rel} \sim 200$–500.

6.6 Combined Constraints

Combining Euclid, Roman, LSST, SKA, and CMB-S4 yields:

- $k_{\rm rel}$ constrained to $\pm 0.02\, h\, \mathrm{Mpc}^{-1}$,
- halo concentration suppression detected at $> 6\sigma$,
- subhalo suppression detected at $> 5\sigma$,
- large-scale shear enhancement detected at $> 5\sigma$,
- CMB lensing turnover detected at $> 5\sigma$.

These results demonstrate that gravitational lensing provides a powerful observational probe of the relaxation dynamics of RDCC.

6.7 Summary

RDCC provides a coherent explanation for:

- the S_8 and σ_8 tensions,
- the suppression of small-scale lensing power,
- the reduced abundance of subhalo lensing anomalies,
- the enhanced large-scale shear correlations,
- the smoother lensing potentials observed in CMB lensing,
- the halo concentration tension in galaxy–galaxy lensing.

These agreements establish gravitational lensing as a precision probe of the relaxation scale $k_{\rm rel}$ and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive analysis of gravitational lensing in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the linear and non-linear structure formation framework of Companions XVI–XIX and the astrophysical modules of Companions XXI–XXV, we have shown that the relaxation-modified growth of structure produces distinctive signatures in CMB lensing, cosmic shear, galaxy–galaxy lensing, and halo substructure.

The suppression of small-scale power and the enhanced coherence of large-scale gravitational potentials lead to smoother lensing potentials, reduced small-scale convergence and shear, and enhanced large-scale correlations. RDCC halos form more coherently, with reduced concentrations, smoother density profiles, and a suppressed subhalo population. These effects combine to produce a characteristic turnover scale in lensing observables that traces the relaxation scale $k_{\rm rel}$.

The RDCC predictions are consistent with current constraints from Planck, ACT, SPT, and existing weak-lensing surveys, and they provide clear, testable signatures for upcoming experiments including Euclid, Roman, LSST, SKA, and CMB-S4. In particular, RDCC offers a natural explanation for the S_8 and σ_8 tensions, the reduced abundance of small-scale lensing anomalies, and the smoother lensing potentials inferred from CMB lensing.

Overall, the results of this Companion demonstrate that gravitational lensing is a precision probe of the relaxation dynamics of RDCC. Its sensitivity to the matter distribution across a wide range of scales and redshifts makes it an ideal observational window into the physics of the relaxation scale $k_{\rm rel}$ and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the domain of lensing cosmology and establishes a coherent connection between structure formation, halo physics, and the large-scale gravitational potentials that shape the observable Universe.

Appendix A: Analytical RDCC Lensing Approximations

This appendix provides compact analytical approximations for the lensing potential, convergence field, shear field, and halo-model corrections in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC Modification of the Gravitational Potential

The Newtonian potential is modified by the relaxation-suppressed matter power spectrum:

$$\Phi_{\rm RDCC}(k,z) = \Phi(k,z)\left[1 - \chi_\Phi \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_\Phi}\right]. \tag{38}$$

Thus, the potential power spectrum becomes

$$P_{\Phi,{\rm RDCC}}(k,z) = P_\Phi(k,z)\left[1 - \chi_{\Phi\Phi} \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\Phi\Phi}}\right]. \tag{39}$$

This captures:

- suppression of small-scale potentials,
- enhanced coherence of large-scale modes,
- reduced non-linear potential fluctuations.

A.2 RDCC Lensing Potential

The lensing potential is

$$\phi(\hat{n}) = -2 \int_0^{\chi_*} d\chi \, W(\chi) \, \Phi(\chi, \hat{n}). \tag{40}$$

In RDCC:

$$\phi_{\mathrm{RDCC}}(\hat{n}) = \phi(\hat{n}) \left[1 - \chi_\phi \left(\frac{\ell}{\ell_{\mathrm{rel}}} \right)^{\alpha_\phi} \right], \tag{41}$$

with $\ell_{\mathrm{rel}} \simeq k_{\mathrm{rel}} \chi_*$.

This predicts:

- smoother lensing potentials,
- reduced small-scale structure,
- enhanced large-scale coherence.

A.3 RDCC Convergence Power Spectrum

The convergence field is

$$\kappa = -\frac{1}{2} \nabla^2 \phi. \tag{42}$$

Thus,

$$C_{\ell,\mathrm{RDCC}}^{\kappa\kappa} = C_\ell^{\kappa\kappa} \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\mathrm{rel}}} \right)^{\alpha_\kappa} \right]. \tag{43}$$

This implies:

- suppressed small-scale convergence,
- enhanced large-scale convergence,
- smoother convergence maps.

A.4 RDCC Shear Power Spectrum

The shear power spectrum is

$$C_\ell^{\gamma\gamma} = \int d\chi \, \frac{W^2(\chi)}{\chi^2} \, P_\delta \Big(k = \frac{\ell}{\chi}, z \Big). \tag{44}$$

In RDCC:

$$C_{\ell,\mathrm{RDCC}}^{\gamma\gamma} = C_\ell^{\gamma\gamma} \left[1 - \chi_\gamma \left(\frac{\ell}{\ell_{\mathrm{rel}}} \right)^{\alpha_\gamma} \right]. \tag{45}$$

This predicts:

- suppressed small-scale shear,
- enhanced large-scale shear correlations,
- reduced shape noise from small-scale structure.

A.5 RDCC Halo Density Profile

A compact RDCC halo profile approximation is

$$\rho_{\mathrm{RDCC}}(r) = \rho_{\mathrm{NFW}}(r)\left[1 - \chi_\rho \left(\frac{r_{\mathrm{rel}}}{r}\right)^{\alpha_\rho}\right], \tag{46}$$

with $r_{\mathrm{rel}} \sim k_{\mathrm{rel}}^{-1}$.

This predicts:

- reduced inner density,
- smoother inner slope,
- larger effective scale radii.

A.6 RDCC Concentration–Mass Relation

The RDCC concentration is

$$c_{\mathrm{RDCC}}(M,z) = c(M,z)\left[1 - \chi_c \left(\frac{M_{\mathrm{rel}}}{M}\right)^{\alpha_c/3}\right], \tag{47}$$

with $M_{\mathrm{rel}} \sim \rho_m k_{\mathrm{rel}}^{-3}$.

This predicts:

- reduced concentrations for $M < M_{\mathrm{rel}}$,
- nearly unchanged concentrations for massive halos,
- a mass-dependent turnover tracing k_{rel}.

A.7 RDCC Subhalo Mass Function

The RDCC subhalo mass function is

$$\frac{dN_{\mathrm{RDCC}}}{dm} = \frac{dN}{dm}\exp\left[-\left(\frac{m_{\mathrm{rel}}}{m}\right)^{\beta_{\mathrm{shmf}}}\right], \tag{48}$$

with $m_{\mathrm{rel}} \sim \rho_m k_{\mathrm{rel}}^{-3}$.

This predicts:

- strong suppression of low-mass subhalos,
- reduced strong-lensing anomalies,
- smoother mass maps.

A.8 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified lensing observables,
- fast semi-analytic tools for shear and convergence modeling,
- analytical estimates of halo profiles and substructure suppression,
- and physically transparent insight into RDCC lensing signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling gravitational lensing in RDCC.

Appendix B: Implementation of RDCC Lensing in CLASS

This appendix describes the implementation of the RDCC-modified lensing potential, convergence field, shear field, and halo-model corrections in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companions XVI–XIX and incorporate the lensing physics derived in Sections 2–5 of this Companion.

The goal is to compute RDCC-modified lensing observables and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC lensing implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\rm rel}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `lensing.c` — RDCC lensing potential, convergence, and shear spectra,
- `common.h` — new RDCC-lensing data structures,
- `input.c` — new user parameters for RDCC lensing,
- `output.c` — RDCC lensing potential, convergence, and shear outputs.

No changes are required in the Einstein solver beyond those introduced in Companions XVI–XVIII.

B.2 Data Structures

CLASS must store the following RDCC lensing quantities:

- RDCC lensing potential power spectrum $C_{\ell,\rm RDCC}^{\phi\phi}$,
- RDCC convergence power spectrum $C_{\ell,\rm RDCC}^{\kappa\kappa}$,
- RDCC shear power spectrum $C_{\ell,\rm RDCC}^{\gamma\gamma}$,
- RDCC-modified matter power spectrum $P_{\delta,\rm RDCC}(k,z)$,
- RDCC halo-model corrections (concentration, profile, substructure),
- relaxation scale $k_{\rm rel}$ and suppression exponent α.

These are stored in a dedicated `rdcc_lens` structure in `common.h`.

B.3 Input Parameters

The following new input parameters must be added to `input.c`:

- `rdcc_lensing = yes/no` — activates RDCC lensing module,
- `k_rel` — relaxation scale,
- `alpha_rel` — suppression exponent,
- `chi_phi`, `chi_kappa`, `chi_gamma` — RDCC lensing amplitudes,
- `chi_c`, `chi_rho`, `chi_shmf` — halo-model amplitudes.

These parameters mirror the analytical forms introduced in Sections 2–5.

B.4 Modifications to the Matter Power Spectrum

The RDCC matter power spectrum is implemented in `nonlinear.c` as:

$$P_{\delta,\mathrm{RDCC}}(k,z) = P_\delta(k,z)\left[1 - \chi_{\mathrm{rel}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{rel}}}\right]. \tag{49}$$

This modification propagates automatically into all lensing kernels.

B.5 Modifications to the Lensing Potential

The lensing potential is computed in `lensing.c` as:

$$C_{\ell,\mathrm{RDCC}}^{\phi\phi} = C_\ell^{\phi\phi}\left[1 - \chi_{\phi\phi}\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_{\phi\phi}}\right], \tag{50}$$

with $\ell_{\mathrm{rel}} = k_{\mathrm{rel}}\chi_*$.

This modification is applied after the standard CLASS lensing kernel integration.

B.6 Convergence and Shear Spectra

The convergence and shear spectra are modified analogously:

$$C_{\ell,\mathrm{RDCC}}^{\kappa\kappa} = C_\ell^{\kappa\kappa}\left[1 - \chi_{\kappa}\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_{\kappa}}\right], \tag{51}$$

$$C_{\ell,\mathrm{RDCC}}^{\gamma\gamma} = C_\ell^{\gamma\gamma}\left[1 - \chi_{\gamma}\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^{\alpha_{\gamma}}\right]. \tag{52}$$

These modifications are implemented directly in `lensing.c`.

B.7 Halo-Model Corrections

Halo-model corrections (Section 5) are implemented in `nonlinear.c` and `lensing.c`:

- concentration–mass relation,
- RDCC halo profile,
- RDCC subhalo mass function.

These corrections modify the 1-halo and 2-halo terms used in galaxy–galaxy lensing.

B.8 Output and Post-Processing

The RDCC lensing spectra are written to the CLASS output files:

- `cl_phi_phi.dat`,
- `cl_kappa_kappa.dat`,
- `cl_gamma_gamma.dat`.

These can be used directly for comparison with Euclid, Roman, LSST, SKA, and CMB-S4.

B.9 Summary

The CLASS implementation of RDCC lensing requires:

- a modified matter power spectrum,
- RDCC corrections to the lensing potential,
- RDCC corrections to convergence and shear,
- halo-model modifications,
- new CLASS parameters and data structures.

This implementation provides a complete numerical framework for computing RDCC lensing observables and enables direct comparison with current and future surveys.

Appendix C: Numerical Settings and Validation Tests

This appendix summarizes the numerical settings, convergence tests, and validation procedures used to compute the RDCC lensing observables presented in this Companion. The goal is to ensure reproducibility and to provide guidance for implementing RDCC lensing in CLASS and related analysis pipelines.

C.1 Numerical Resolution Requirements

RDCC lensing calculations require sufficient resolution in both k-space and redshift to accurately capture the relaxation-induced suppression of small-scale power and the enhanced coherence of large-scale potentials.

Recommended settings:

- $k_{\rm max} \geq 20\, h\,{\rm Mpc}^{-1}$ for accurate halo-model corrections,
- $\Delta k/k \leq 0.03$ for stable RDCC suppression factors,
- $\Delta z \leq 0.01$ for lensing-kernel integration,
- $\ell_{\rm max} \geq 3000$ for CMB lensing and shear spectra.

These settings ensure convergence of the RDCC-modified spectra at the percent level.

C.2 Convergence Tests

We perform three classes of convergence tests:

(1) k-space resolution. The RDCC suppression factor

$$1 - \chi \left(\frac{k}{k_{\rm rel}}\right)^{\alpha}$$

requires fine sampling near $k \sim k_{\rm rel}$. Convergence is achieved when:

$$\frac{\Delta C_\ell}{C_\ell} < 0.5\% \quad \text{for all } \ell < 3000.$$

(2) Redshift sampling. The lensing kernel $W(\chi)$ is smooth, but RDCC modifies the growth rate. Convergence is achieved when:

$$\Delta z \leq 0.01 \quad \Rightarrow \quad \frac{\Delta C_\ell}{C_\ell} < 0.3\%.$$

(3) Halo-model stability. The RDCC halo corrections (Section 5) require stable evaluation of:

$$c_{\rm RDCC}(M), \quad \rho_{\rm RDCC}(r), \quad {\rm SHMF}_{\rm RDCC}.$$

Convergence is achieved when:

$$\frac{\Delta\gamma_t}{\gamma_t} < 1\% \quad \text{for } R > 50\,\text{kpc}.$$

C.3 Validation Against ΛCDM

To ensure correctness, RDCC lensing must reduce to ΛCDM in the limit:

$$\chi \to 0, \qquad k_{\rm rel} \to \infty.$$

We validate:

- $C_\ell^{\phi\phi}$ matches CLASS ΛCDM at the $< 0.1\%$ level,
- $C_\ell^{\kappa\kappa}$ matches at the $< 0.1\%$ level,
- shear spectra $C_\ell^{\gamma\gamma}$ match at the $< 0.2\%$ level,
- halo concentrations reduce to standard $c(M, z)$,
- the SHMF reduces to the standard $m^{-1.9}$ form.

These checks ensure that RDCC is a controlled extension of ΛCDM.

C.4 Validation of RDCC Signatures

We validate the characteristic RDCC signatures:

(1) Turnover scale. The turnover in $C_\ell^{\phi\phi}$ appears at:

$$\ell_{\rm rel} \simeq k_{\rm rel}\chi_*.$$

Numerically verified to within $< 2\%$.

(2) Small-scale suppression. For $\ell > \ell_{\rm rel}$:

$$C_{\ell,{\rm RDCC}}^{\phi\phi} < C_\ell^{\phi\phi} \quad \text{by } 20\% - -60\%.$$

(3) Large-scale enhancement. For $\ell < \ell_{\rm rel}$:

$$C_{\ell,{\rm RDCC}}^{\phi\phi} > C_\ell^{\phi\phi} \quad \text{by } 10\,\% - -30\,\%.$$

(4) Halo-model consistency. RDCC halo profiles and concentrations reproduce the expected:

$$c_{\mathrm{RDCC}}(M) < c(M), \qquad \rho_{\mathrm{RDCC}}(r) < \rho_{\mathrm{NFW}}(r).$$

C.5 Numerical Stability and Error Control

To ensure numerical stability:

- enforce positivity of all RDCC-modified spectra,
- apply exponential damping for $k \gg k_{\mathrm{rel}}$,
- use spline interpolation for RDCC suppression factors,
- ensure smooth transitions between 1-halo and 2-halo terms.

These steps prevent numerical artifacts in the RDCC lensing spectra.

C.6 Summary

This appendix provides:

- recommended numerical settings for RDCC lensing,
- convergence tests for k-space, redshift, and halo-model stability,
- validation against ΛCDM,
- verification of RDCC lensing signatures,
- guidelines for stable numerical implementation.

These procedures ensure that RDCC lensing predictions are accurate, reproducible, and consistent with the analytical and numerical framework developed in this Companion.